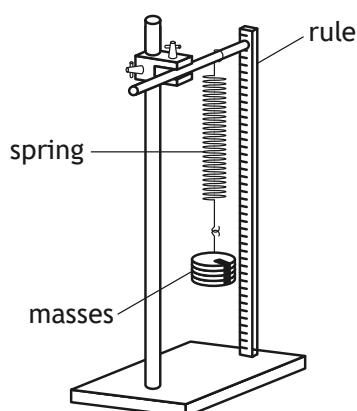


16. A group of students use the apparatus shown to carry out an investigation into Hooke's Law and find the spring constant k of a spring.



The spring constant k is a measure of the stiffness of a spring.

Hooke's Law states that the increase in the length of a spring is proportional to the force used to stretch the spring:

$$F = ke$$

where: F is the force used to stretch the spring, in N
 k is the spring constant, in N m^{-1}
 e is the increase in the length of the spring, in m.

The students obtain the following data.

F (N)	e (mm)
1.0	3
2.0	8
2.9	11
3.9	15
4.9	20
5.9	24

- (a) (i) Using the square-ruled paper on *page 50*, draw a graph of F against e . 3
 (The table of results is also shown on *page 51*, opposite the square-ruled paper.)



* X 8 5 7 7 6 0 1 4 8 *

16. (a) (continued)

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- (ii) Use your graph to determine the spring constant k .

2

Space for working and answer

- (iii) Suggest one improvement to the experimental procedure that the students could make to improve the accuracy of their final result.

1

- (b) When a spring is extended, work is done. Elastic potential energy is stored in the spring. If it is not over-stretched, the elastic potential energy is equal to the work done.

The elastic potential energy can be determined using the relationship

$$E_e = \frac{ke^2}{2}$$

where: E_e is the elastic potential energy, in J

k is the spring constant, in N m^{-1}

e is the increase in the length of the string, in m.

Determine the elastic potential energy stored when the spring extension is 22 mm.

2

Space for working and answer

[END OF QUESTION PAPER]



* X 8 5 7 7 6 0 1 4 9 *

